Does dependency-length influence expectancy in reading times and ERPs?

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A wide range of behavioral and neurophysiological findings support expectation-based models of sentence processing, showing that words that are more expected require less processing effort (Hale, 2001, 2006; Levy, 2008). However, other research has shown long-distance dependencies reliably increase processing difficulty due to working-memory constraints (see Futrell et al., 2020, for discussion). More recent models such as Lossy-Context Surprisal (LCS) integrate both perspectives by deriving surprisal from lossy, imperfect memory representations (Futrell et al., 2020; Hahn et al., 2022). We test the prediction that increased dependency length will modulate expectation effects as a result of lossy memory representations, leading to reduced facilitation for expected words and reduced difficulty for unexpected words when parts of the predictive context are lost over longer distances.

To investigate the interaction of expectation and dependency length, we conducted both a self-paced reading study and an EEG study in German using a 2×2 design based on materials adapted from Aurnhammer et al. (2021), both of which had comprehension questions on every trial. We created 120 items that varied dependency length between the main verb (e.g. *sharpened*) and the object (*axe*) by manipulating the linearization of unassociated elements (*who watched the news … was*) such that the unassociated material either preceded the subject or appeared between the verb and the object. The expectancy of the object (Expected vs. Unexpected) was manipulated by the main verb in the preceding context (*sharpened the axe* vs. *weaved the axe*, see Table 1). Cloze pretests and surprisal values confirmed that the expectancy differences were not affected by the linearization manipulation. Under the LCS account, we should observe an interaction between expectancy and distance such that surprisal effects should be stronger in the short-distance conditions as compared to long-distance conditions.

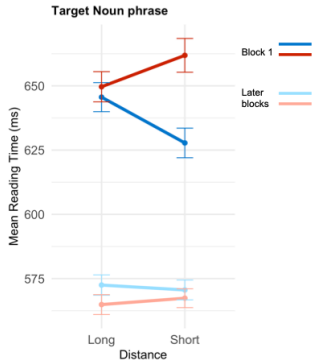
In the on-line reading-time study (N=80), we find the main effect of expectancy and the interaction in the target noun phrase region in the first out of three blocks of the experiment (see Figure 1), consistent with the predictions of LCS. The absence of effects in later blocks is likely due to participants' overall speed-up across the experiment and possibly increasing familiarity with the comprehension questions over the course of the experiment.

In an ERP study (SOA: 500 ms) using the same stimuli, but with more demanding comprehension questions, we observed additive effects of expectancy and distance in the N400 time-window (300–500 ms; see Graphs 2a and 2b) for the target noun phrase, increasing from parietal to frontal regions. The findings suggest processing difficulty is similarly increased by both unexpectedness and the greater memory demands in the long-distance conditions. No interaction, as predicted by LCS, is observed. In the later time-window (500–1000 ms), we observe a significant sustained negativity for long-distance conditions in line with previous findings (Bornkessel et al., 2004, Schacht et al., 2014).

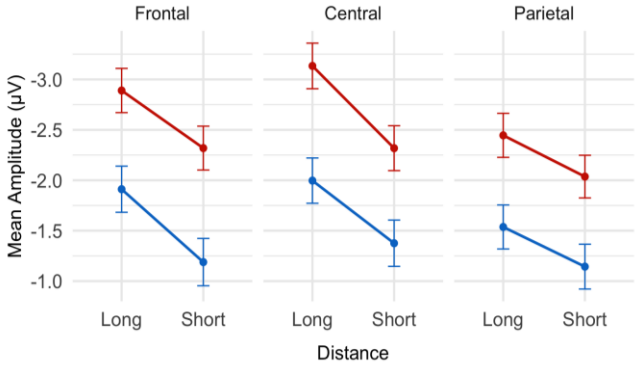
While both studies find effects of distance and expectancy, only the reading times revealed the interaction predicted by LCS. The absence of an interaction in the N400 – driven by an across the board influence of distance – may be due to differences in the presentation and task, or also reflect differential sensitivity of ERPs to memory demands. In sum, while evidence from reading times provided evidence that increased memory demands modulate surprisal in human comprehension as predicted specifically by LCS, the N400 findings indicate that expectancy and memory constraints similarly influence the processing effort indexed by this component.

Conditions	
A Expected Short	Gestern nach dem Hören der Nachrichten und des Wetterberichts **schärfte** der Holzfäller, der ungeduldig war, die **Axt** in seinem… *(Yesterday after listening to the news and the weather report **sharpened** the lumberjack, who impatient was, the **axe** in his…)*
B Expected Long	Gestern **schärfte** der Holzfäller, der die Nachrichten und den Wetterbericht gehört hatte und ungeduldig war, die **Axt** in seinem… *(Yesterday **sharpened** the lumberjack, who the news and the weather report listened had and impatient was, the **axe** in his…)*
C Unexpected Short	Gestern nach dem Hören der Nachrichten und des Wetterberichts **webte** der Holzfäller, der ungeduldig war, die **Axt** in seinem… *(Yesterday after listening to the news and the weather report **weaved** the lumberjack, who impatient was, the **axe** in his…)*
D Unexpected Long	Gestern **webte** der Holzfäller, der die Nachrichten und den Wetterbericht gehört hatte und ungeduldig war, die **Axt** in seinem… *(Yesterday **weaved** the lumberjack, who the news and the weather report listened had and impatient was, the **axe** in his…)*

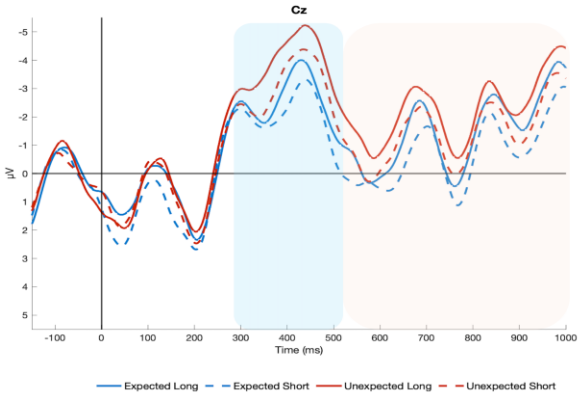
Table1. Example of stimuli.



Graph1. Mean reading times of target NP



Graph 2a. Mean amplitude of N400 time window



Graph 2b. Averaged ERP waveforms (Cz)

Selected References

Aurnhammer et al., (2021), *PLOS ONE*; Bornkessel et al., (2004), *Cognitive Brain Research*; Futrell et al., (2020), *Cognit. Sci.*; Hahn et al., (2022), *PNAS.*; Hale, J. (2001), *North American Chapter of the Association for Computational Linguistics*; Hale, J. T. (2016), *Language and Linguistics Compass*; Levy, R. (2008), *Cognition*; Schacht et al., (2014), *PLOS ONE*;